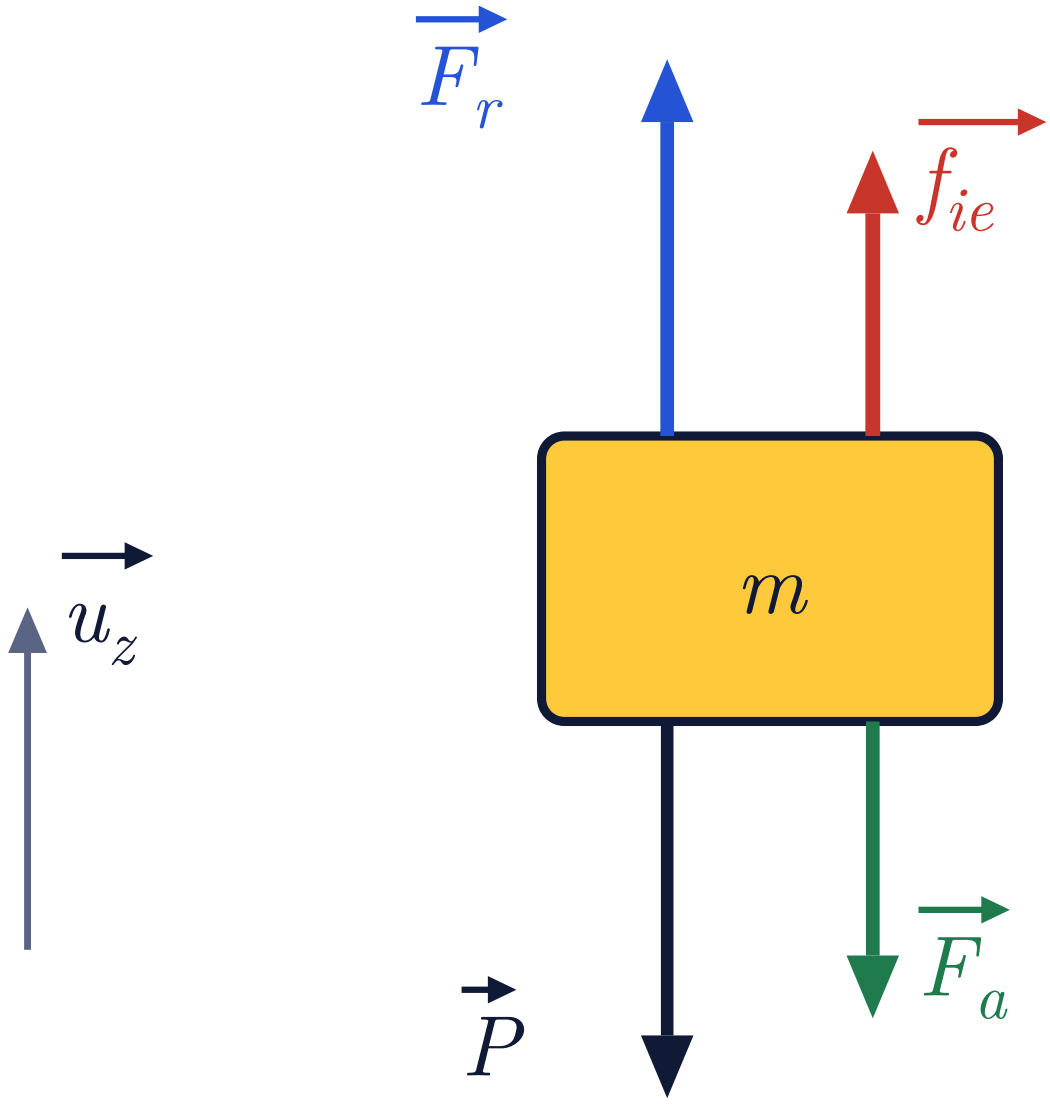
$\vec{F}_r$
$\vec{f}_{ie}$
m
$\vec{u}_z$
$\vec{P}$
$\vec{F}_a$